

# Auditing a Transcriptomic Prognostic Funnel in Clear Cell Renal Cell Carcinoma

HYDRA-ccRCC development manuscript

**Internal development draft. Not for direct submission.**

## Abstract

The one-gene HYDRA funnel fails its prespecified held-out test, while the broader reproducible gene set still carries ranking information. On one TCGA-KIRC tumor per patient, the selection-aware ten-repeat, five-fold analysis changed concordance by 0.0039 (patient-bootstrap 95% interval -0.0088 to 0.0171, conditional on the saved out-of-fold predictions). A training-only ridge Cox model of the reproducible genes on those same splits changed concordance by 0.0257 (interval 0.0116 to 0.0413). The ridge penalty was chosen inside each training fold by partial-likelihood deviance; the median selected  $\lambda$  was 40.22. That arm was not the acceptance test, and its three-year Brier interval still crossed zero. Survival-only and differential-expression-only single genes stayed within about 0.002 of the clinical concordance. The corrected funnel, from 541 aliquots to 533 patients, retained 3,323 reproducible differentially expressed genes, 538 strict candidates, and 23 high-confidence candidates. A published 34-gene score on the same splits changed concordance by 0.0129 (interval -0.0018 to 0.0287). Alternate aliquot rules and an unpaired GSE53757 fit did not replace the primary shortlist. A matched-list comparison did not establish superiority over survival-only selection (GSE29609 directional-rate difference 0.042; 95% bootstrap interval -0.194 to 0.270). E-MTAB-1980 gave strict same-direction support to 12 of 22 mapped candidates, while GSE29609 retained the TCGA direction for only 5 of 21 and showed FDR-significant reversals for DDC and TCIRG1. No candidate-by-treatment interaction survived FDR correction in the CheckMate 025 RNA subset. These are candidate prognostic associations and audit findings, not a validated biomarker panel.

## 1 Introduction

Clear cell renal cell carcinoma (ccRCC) has broad metabolic, vascular, and immune transcriptional changes [4, 12]. A tumor-normal expression difference does not establish independent prognostic value: a signal can arise from the composition of the tissue, and repeated samples from one patient can make survival evidence appear more precise. We audited the public-data HYDRA-ccRCC funnel for these failure modes. The question is whether the one-gene rule generalizes, and whether the broader reproducible set still ranks patients when that rule does not.

## 2 Methods

### 2.1 Inputs, patient identity, and discovery

TCGA-KIRC STAR unstranded counts and clinical outcomes were retrieved through TCGAbiolinks [5]. For each patient and sample type, we selected the aliquot with greatest raw-count library depth and broke barcode ties so the choice is deterministic rather than biological. The 541 tumor aliquots came from 533 patients, including 175 deaths: four patients contributed three aliquots each, which accounts for all eight extra aliquots. All 72 normals were already unique by patient. Genes are matched by symbol after a dataset-specific collapse, recorded in the gene-mapping table. TCGA strips the Ensembl version, maps the id with org.Hs.eg.db, and keeps the first symbol. GSE40435, GSE53757, and GSE29609 take the first token of the platform gene-symbol field and average probes that share it. E-MTAB-1980 maps RefSeq identifiers the same way and averages rows that share a symbol. TRACERx matches candidate symbols to expression rownames.

CheckMate 025 averages rows that share `gene_name`. Survival-event coding was checked against vital status and the appropriate death or follow-up date; joins were required to remain one row per patient. Three alternate gates, reported as sensitivities, dropped those four patients, kept the first barcode, or summed a patient’s tumor counts. Original retrieval dates for cached public inputs were unavailable, so the repository records source URLs and hashes without assigning new access dates.

DESeq2 compared the selected tumors and normals after requiring at least ten counts in ten samples [9]. A separate patient-blocked DESeq2 fit, with design patient plus tissue type, used the 72 selected tumor-normal pairs. GSE40435 and GSE53757 were analyzed with patient-blocked limma; each parsed patient was required to have exactly one tumor and one normal, and surrogate-variable analysis protected the paired contrast [8, 10]. GSE53757 has no field that stores one shared patient id for every pair. Adjacent rows were opposite tissues and the same stage in all 72 pairs, and 4 pairs also shared a title token, so the primary pair ids follow that adjacent order. An unpaired limma fit is a sensitivity. A reproducible DEG required TCGA FDR below 0.05 and absolute  $\log_2$  fold change at least one, the same sign in both GEO cohorts, and nominal support in at least one, defined as a two-sided unadjusted  $p < 0.05$  with that same sign. A separate apegglm analysis tested all count-QC genes without the fold-change gate; it was not allowed to redefine the primary candidate set [13].

The primary Cox model used continuous standardized expression, age, sex, categorical AJCC stage with Stage I as the reference, and grade collapsed to G1+G2 versus G3+G4 among complete cases [6]. G1 contains 14 patients in the TCGA clinical table, so it was not kept as its own level. The strict gate also required main-model FDR below 0.05, absolute log hazard ratio at least  $\log(1.25)$ , both GEO effects at least 0.25 in absolute  $\log_2$  units, and concordant nominal support in two sensitivity fits: expression plus age, sex, and stage, and expression plus age, sex, and grade. The high-confidence gate required FDR below 0.01 and absolute log hazard ratio at least  $\log(1.5)$ . Those magnitude cutoffs are prespecified project thresholds, not a cited minimal clinically important difference. False-discovery correction stays inside one family at a time: differential expression, each GEO cohort, Cox models among reproducible genes, and the high-confidence tests for shape, composition, and treatment interaction. Proportional-hazards tests were diagnostics, never selection gates. The selected gene is the one with the highest evidence score,  $-\log_{10}(\text{FDR}) + |\log \text{HR}|$  plus bounded fold-change terms, with remaining ties broken by FDR and then gene id. The acceptance test adds that one gene. The ridge fit is a separate multigene description. It is a glmnet Cox model with clinical covariates unpenalized and an L2 penalty on every training-fold reproducible gene.  $\lambda$  minimizes inner-fold partial-likelihood deviance on training patients only. glmnet standardizes the training columns and applies those moments to the held-out rows. The penalty does not zero coefficients. This fit does not replace the one-gene test.

## 2.2 Selection-aware prediction and null simulation

Ten repeats of five event-stratified patient folds reran TCGA DE and every candidate gate within each training fold of the common complete-case clinical population. Fold labels were redrawn for each repeat. Event stratification shuffled deaths and censored patients separately, and every outer test fold contained 34 deaths. The full-data stage and grade sensitivities include some additional patients and are therefore not numerically identical to these nested fits. GEO expression evidence remained fixed; test patients and their normals were excluded from training. That is the scope of “selection-aware” here: TCGA differential expression, the fixed GEO gate, the Cox gates, and the choice of one gene. It does not re-split the GEO cohorts or re-choose the 0.01 rule inside a fold. Library-size normalization and expression scaling were estimated on training samples and applied to held-out tumors. The nested fit used  $\log_2$  normalized counts, while the full-data survival analysis used a variance-stabilizing transformation; this preserves training-only preprocessing but changes the expression scale, limiting direct coefficient comparisons. DESeq2 retained the primary outlier-replacement setting, with a logged no-replacement retry only if the pinned runtime failed in that step. The top qualifying gene was added to a clinical Cox model; when no gene qualified, its prediction equaled the clinical-only prediction. Out-of-fold concordance, three-year inverse-probability-of-censoring-weighted Brier score, and risk-score calibration slope were computed per repeat. The concordance increment received a patient-resampled 95% interval conditional on the saved out-of-fold predictions; it does not include variation from refitting selection or redrawing folds. The prespecified success rule required an increment at least 0.01, a conditional interval above zero, and a mean Brier difference no greater than zero. The 0.01

increment is this project’s size requirement, not a universal clinical standard. A lower Brier score is not, by itself, better calibration: discrimination and calibration contribute separately to the Brier score.

Two hundred clinical-only null datasets used an exponential baseline calibrated to the clinical Cox cumulative hazard with resampled censoring; each reran the full training selection on one held-out fold, cycling across the five fixed folds from repeat 1. This is a parametric clinical-risk null, not another ten-by-five-fold interval and not a p-value for the concordance increment. A second null permuted training survival pairs on the five saved repeat-1 differential-expression fits, without refitting DESeq2, and cycled the same way across 200 simulations. The original 1,000 patient bootstraps and 20-repeat per-gene CV condition on the full-data shortlist and remain descriptive rather than selection-adjusted evidence.

The same patients, seeds, and event-stratified folds were then used to compare selectors. One training-only DESeq2 fit supplied six predictors for each held-out fold: the clinical model; the primary HYDRA gene; a survival-only gene held to the same high-confidence hazard gates without tumor-normal or GEO filters; a differential-expression-only gene held to the training DESeq2 thresholds without survival or GEO filters; a ridge Cox model on every training-fold reproducible DEG, with clinical covariates unpenalized and the penalty chosen by training-only cross-validation [11]; and one expression-matched gene at or below the median residual correlation with the clinical martingale residuals. Survival-only false-discovery rates were computed as if every unscreened count-QC gene had  $p = 1$ , after fitting the top 2,000 residual correlations. The HYDRA gene was required to match the saved primary nested result. Intervals remain conditional on the saved predictions. This comparison does not replace the prespecified one-gene acceptance rule.

On those same folds we also added ClearCode34, the published 34-gene ccA/ccB score [3]. ccA genes were signed negative and ccB genes positive. Current symbols were taken from gene aliases where the published symbol has changed; one published symbol did not map. Expression was  $\log_2$  counts per million, and the training fold supplied the mean and standard deviation. This arm does not select genes. Separately, on the previously inspected E-MTAB-1980 cohort, we fit in-sample Cox models for the study’s ccA/ccB label, for the highest-evidence primary gene present on that array (CRYL1; CYFIP2 did not map), and for a sign-weighted mean of the mapped high-confidence genes, each added to the limited clinical model. Those E-MTAB fits are not nested, and their in-sample calibration slopes are not held-out estimates.

### 2.3 Exploratory alternatives and biological checks

At the largest common list size of 22 genes, we compared the complete rule with survival-only, DE-only, each leave-one-GEO-out rule, and TCGA-expression-matched non-prognostic controls. GSE29609 and E-MTAB-1980 were previously inspected, so both evaluations are exploratory. We estimated the complete-rule minus survival-only directional-rate difference and a 2,000-draw paired gene bootstrap in each cohort. The prespecified funnel criterion required an interval entirely above zero.

Candidate Cox models were refit on matched complete cases with published TCGA consensus tumor purity [1] or leave-one-candidate-out marker scores. Each score is the mean of sample-standardized marker expression, then standardized across samples. The sets are proximal tubule (AQP1, LRP2, CUBN, SLC5A2, ALDOB), endothelial (PECAM1, VWF, KDR, EMCN, TEK), immune (PTPRC, CD3D, CD8A, MS4A1, CD68), and stromal (COL1A1, COL1A2, DCN, LUM, ACTA2). A candidate that belongs to a set is omitted from that score. These are bulk marker scores, not measured cell proportions. Consensus purity is an inferred bulk estimate. Nonlinear expression and expression-by-log-time models were tested against the linear proportional-hazards fit. Human Protein Atlas normal-tissue single-cell profiles informed cell-source cautions but were not treated as ccRCC tumor single-cell validation.

TRACERx Renal provided primary-region expression and linked survival data from a pinned public repository [7]. One thousand repeats selected one region per patient. A fixed event-stratified 39-patient subset isolated region-choice variation; a separate scenario resampled which 39 patients entered each repeat. Check-Mate 025 supplementary RNA data supported post hoc continuous-expression-by-treatment Cox interactions for overall and progression-free survival, with FDR correction within endpoint and model [2].

## 3 Results

### 3.1 Corrected discovery and conditional sensitivity

The original 541 tumor aliquots represented 533 patients; four patients contributed three aliquots each, accounting for the eight extra aliquots. No patient had more than one normal. Patient-level selection left 72 normals. Dropping the four multi-aliquot patients left 19 high-confidence genes, the first-barcode rule left 20, and summing tumor counts left 19. GRAMD1A, IFFO1, LTB4R, and RBM47 were absent from all three alternate sets. The deepest-library rule stays the primary rule. The unpaired GSE53757 gate kept 23 of the 23 primary genes and had Jaccard overlap 0.998 with the paired reproducible list. The paired DE fit included 72 patients and found 8,626 genes meeting the stated DE threshold. All 23 candidates retained their tumor-normal direction, while only 17 met the paired DE threshold; this is a sensitivity to the tumor-normal design, not an independent cohort. The primary analysis found 8,534 TCGA-significant mapped genes, 3,323 reproducible DEGs, 1,186 main-model FDR associations, 1,117 sensitivity-pass genes, 538 strict candidates, and 23 high-confidence candidates. Relative to the previous 27-gene run, 3 genes entered and 7 left the shortlist. The gene-level columns for all 23 candidates, including external survival, composition, nested selection frequency, TRACERx direction, and the CheckMate interaction, are in the candidate evidence matrix.

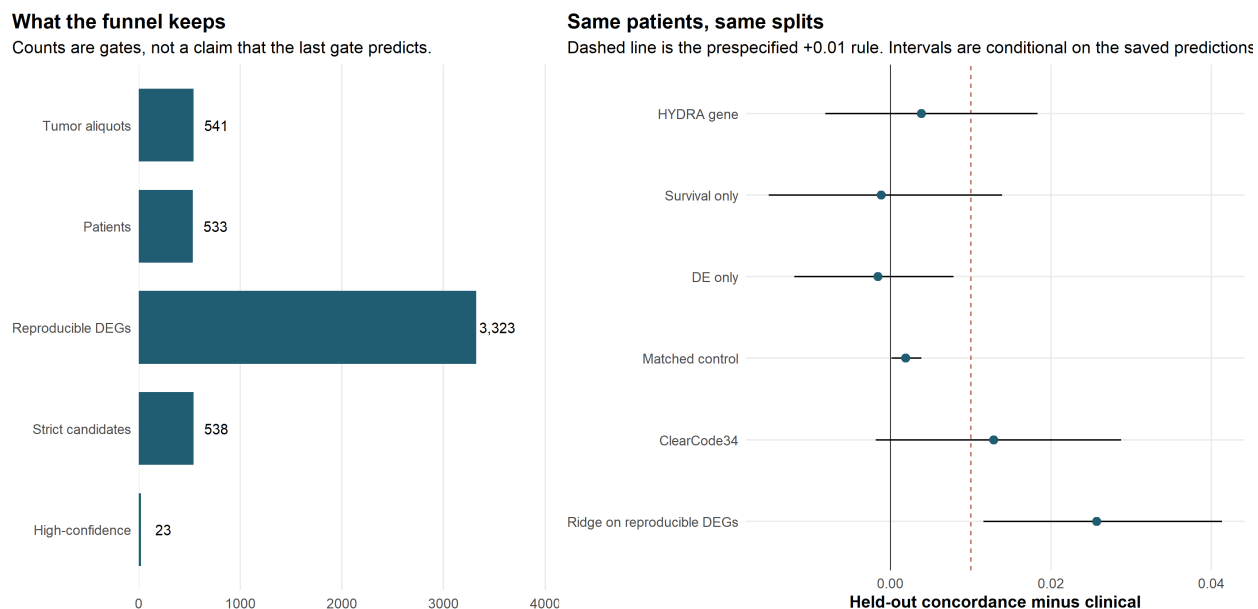


Figure 1: Left: the selection funnel from tumor aliquots to the high-confidence set. Right: held-out concordance minus the clinical model on the same outer splits. The dashed line is the prespecified +0.01 increment. Intervals are conditional on the saved predictions.

The all-gene sensitivity modeled 32,192 count-QC genes and found 12,630 survival associations at global FDR below 0.05, illustrating how broad the outcome signal becomes without the upstream gate.

All 23 conditional bootstrap intervals excluded zero. The per-gene conditional CV showed positive mean concordance increments for 23 candidates, but both analyses start from genes chosen on the full TCGA outcome data. They cannot estimate the optimism of the complete selection process. The within-reproducible-gene overlap permutation did not show enrichment of the GEO-effect gate among survival-pass genes (empirical  $p = 0.178$ ). It shuffles pass labels rather than refitting survival models, so it is a limited negative control.

### GSE53757 pairing sensitivity

All 23 high-confidence genes. Jaccard overlap of the reproducible lists is 0.998.

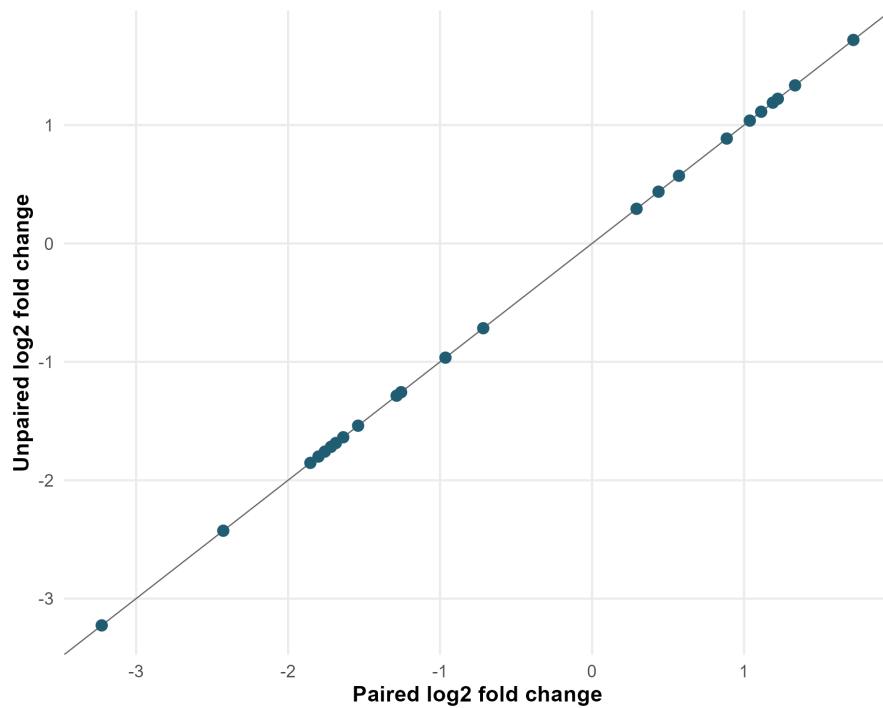


Figure 2: Paired versus unpaired GSE53757  $\log_2$  fold changes for the 23 high-confidence genes. The unpaired gate kept every gene. The reproducible-list Jaccard overlap was 0.998.

### HYDRA-ccRCC Evidence Funnel

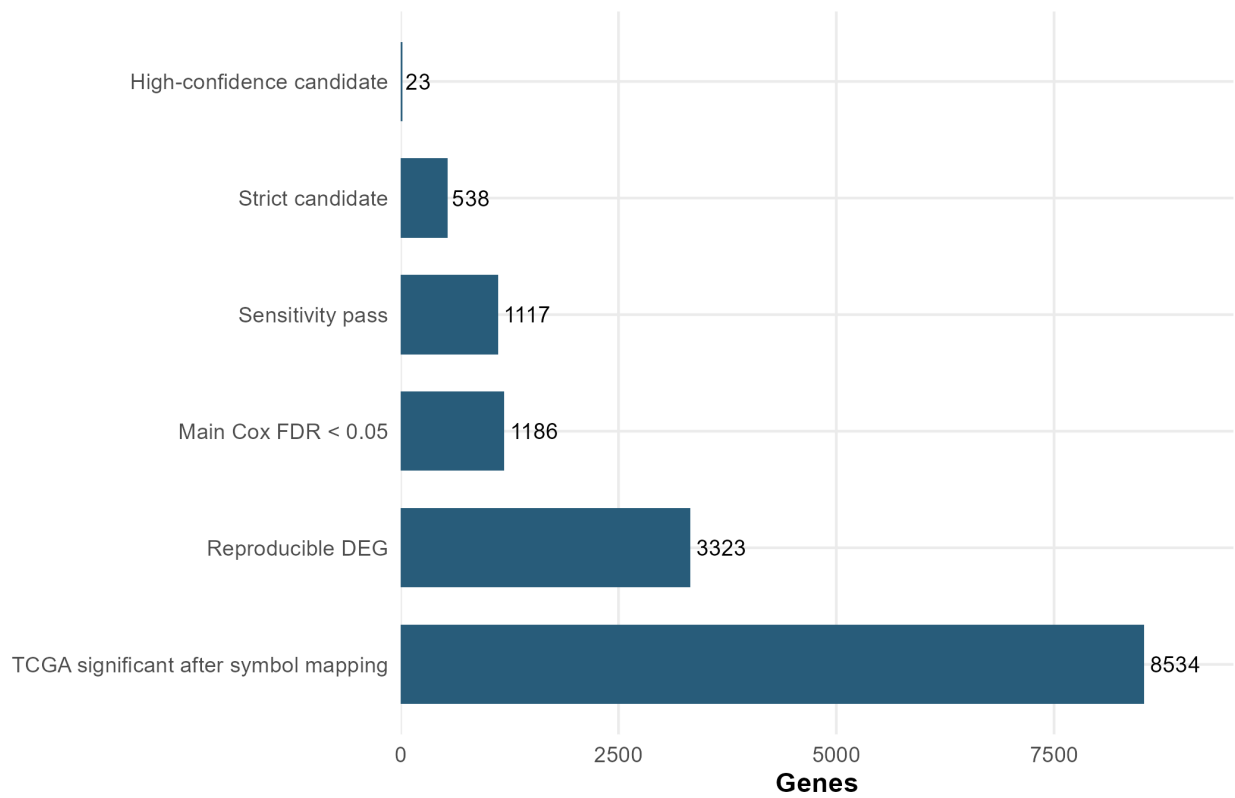


Figure 3: Corrected patient-level discovery funnel. The candidate set is defined entirely within TCGA and the two GEO expression cohorts.

### 3.2 Selection-aware prediction and funnel comparison

The nested analysis selected no gene in 0 of 50 training folds, but selected 10 different genes overall; the most frequent appeared in 22 folds. Thus, universal selection did not imply a stable gene identity.

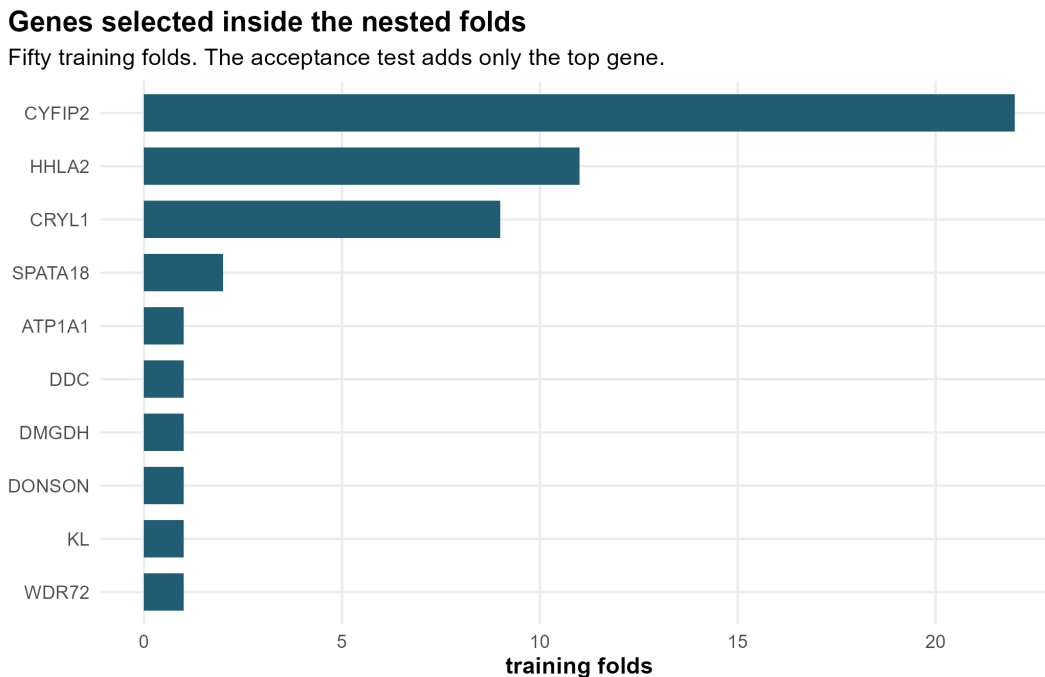


Figure 4: Training-fold selection frequency. Ten genes were chosen at least once. CYFIP2 was the top gene in 22 of 50 folds.

2 folds required a no-replacement retry after a DESeq2 outlier-replacement error. Mean out-of-fold concordance was 0.744 for the clinical model and 0.747 after adding the training-selected gene, an increment of 0.0039 (patient-bootstrap 95% interval -0.0088 to 0.0171). The mean gene-minus-clinical three-year Brier-score difference was -0.0009. Mean risk-score calibration slopes were 0.86 and 0.76, respectively; the latter was farther from one despite the slightly lower average Brier score. The predeclared prediction criterion failed. The clinical-only null selected a gene in 3 of 200 single-fold simulations. The permutation null, which keeps the training expression fits and scrambles training survival, selected a gene in 0 of 200 simulations. Neither frequency is a p-value for the ten-repeat mean increment.

On those splits, the benchmark HYDRA arm matched the primary selected gene in every fold and again changed concordance by 0.0039 (interval -0.0081 to 0.0183). Survival-only selection, which used 20 genes, changed concordance by -0.0011 (interval -0.0151 to 0.0139). Differential-expression-only selection concentrated on 5 genes and changed concordance by -0.0015 (interval -0.0120 to 0.0079). Both left the mean three-year Brier score worse than the clinical model. The expression-matched control, spread across 44 genes, changed concordance by 0.0019 (interval 0.0001 to 0.0039). The interval sits above zero and remains well below 0.01, so a nonzero lower bound at this size is not a useful gain. Ridge regression kept the training-fold reproducible genes, on average 3,335 genes (range 3,067 to 3,447). Concordance rose by 0.0257 (interval 0.0116 to 0.0413), clearing the 0.01 line. The mean Brier difference was -0.0063 (interval -0.0141 to 0.0017), so the probability-score improvement is not stable under patient resampling. Calibration slopes were 0.86 for the clinical model, 0.76 after the single HYDRA gene, and 0.87 after the ridge model. The eligible set therefore carried held-out ranking information that none of the one-gene rules captured. Because the ridge comparison was specified after the one-gene rule, it does not reopen the failed acceptance test. ClearCode34, using 33 mapped genes, changed concordance by 0.0129 (interval -0.0018 to 0.0287) and the three-year Brier score by -0.0046. The calibration slope was 0.83. The interval includes values below zero and below 0.01, so this published score does not meet the prespecified rule either. On E-MTAB-1980, adding the study

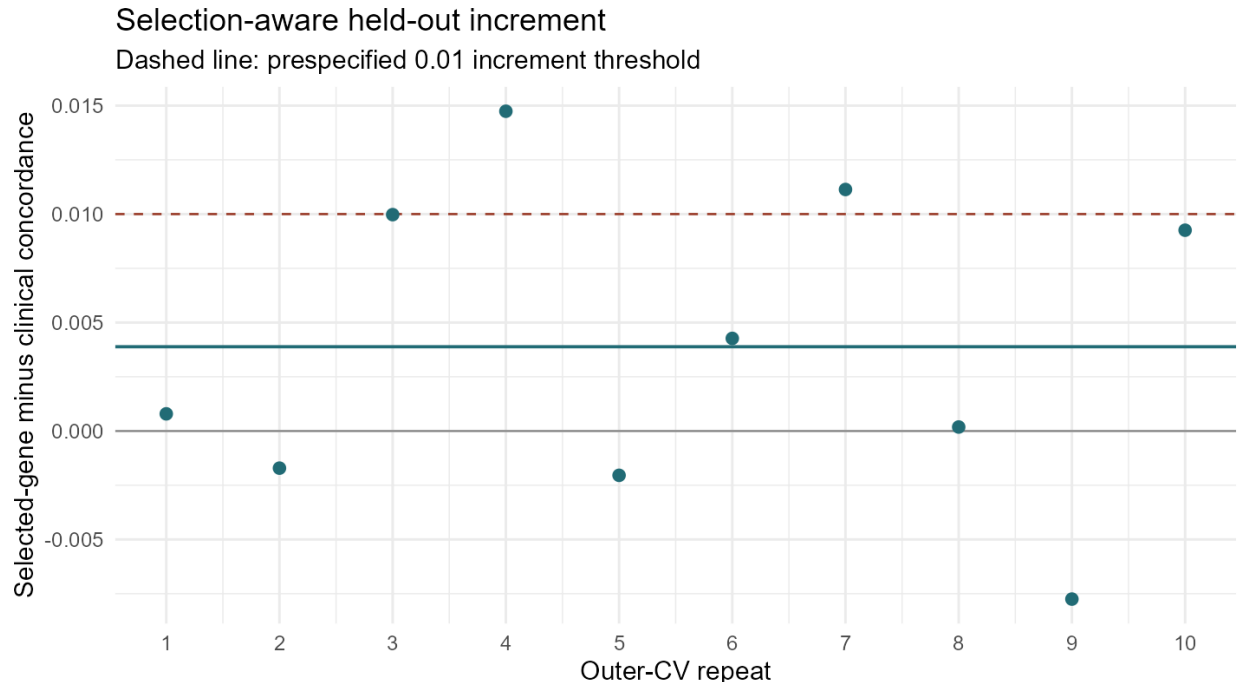


Figure 5: Concordance increment across ten complete outer-CV repeats. The dashed line marks the prespecified 0.01 threshold; the patient-bootstrap interval is reported in text.

ccA/ccB label to the limited clinical model changed in-sample concordance by 0.0351 (interval -0.0483 to 0.1153). Adding CRYL1 changed it by 0.0034, and the signed panel of 22 mapped genes changed it by 0.0314. Those intervals include zero. The cohort has 23 deaths and was previously inspected.

The matched-list funnel criterion failed. In GSE29609 the complete-minus-survival-only directional-rate difference was 0.042 (paired gene-bootstrap 95% interval -0.194 to 0.270). The complete and survival-only lists mapped 20 and 19 genes there, respectively, so the observed difference compares proportions with distinct denominators. In E-MTAB-1980 both lists had 100% direction agreement among mapped genes (21 and 18, respectively), leaving no directional separation. Matching at 22 selected genes was necessary because one leave-one-GEO-out rule yielded only 22 eligible genes. The comparison reuses already inspected external outcomes and is not confirmatory.

### 3.3 External disagreement and composition

ACADM, CRYL1, and DDC are three evidence profiles, not a ranked set of winners. ACADM is a concordant metabolic association: downregulated in TCGA and both GEO cohorts, with a protective TCGA hazard ratio. CRYL1 has a stronger main-model FDR than ACADM and is the high-confidence gene scored on E-MTAB-1980, because the nested procedure’s most frequent gene, CYFIP2, did not map to that array. DDC has a large GEO downregulation and a protective TCGA association, then reverses in GSE29609 at FDR below 0.05. The matrix records the same columns for the other genes. These are candidate associations, not mechanisms or clinical biomarkers.

GSE29609 contained 39 patients and 17 deaths. It mapped 21 primary candidates, of which 5 retained the TCGA hazard direction; 4 had nominal opposite-direction associations. DDC and TCIRG1 were reversed at FDR below 0.05. E-MTAB-1980 contained 101 patients and 23 deaths, mapped 22 candidates, retained the TCGA direction for 21, and gave 12 strict same-direction unadjusted and limited-adjustment FDR results. These cohorts disagree sufficiently that the latter cannot be described as universal external replication.

Matched-patient marker-score adjustment removed FDR support for 6 candidates: CLCN5, DDC, GJB1, HIBCH, KL, PODXL. Direct consensus-purity adjustment retained direction and FDR support for all 23.

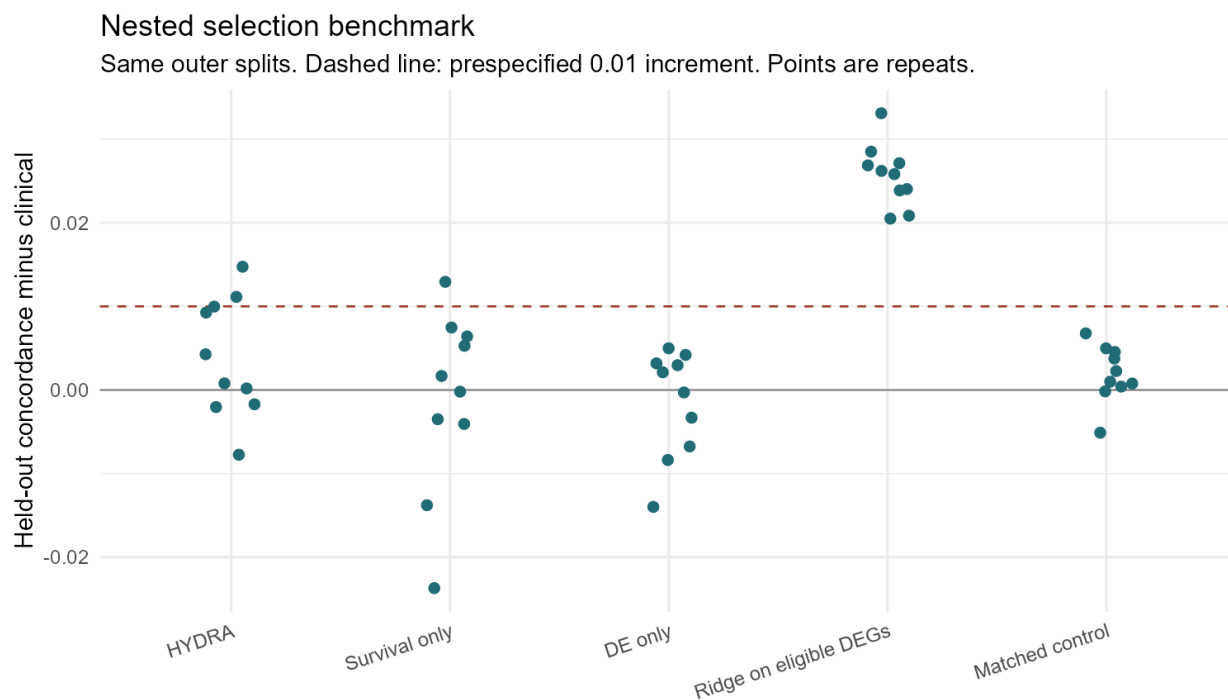


Figure 6: Repeat-level concordance increments for the five selectors fit inside the nested benchmark. ClearCode34 is on the master figure. The dashed line is the prespecified 0.01 threshold.

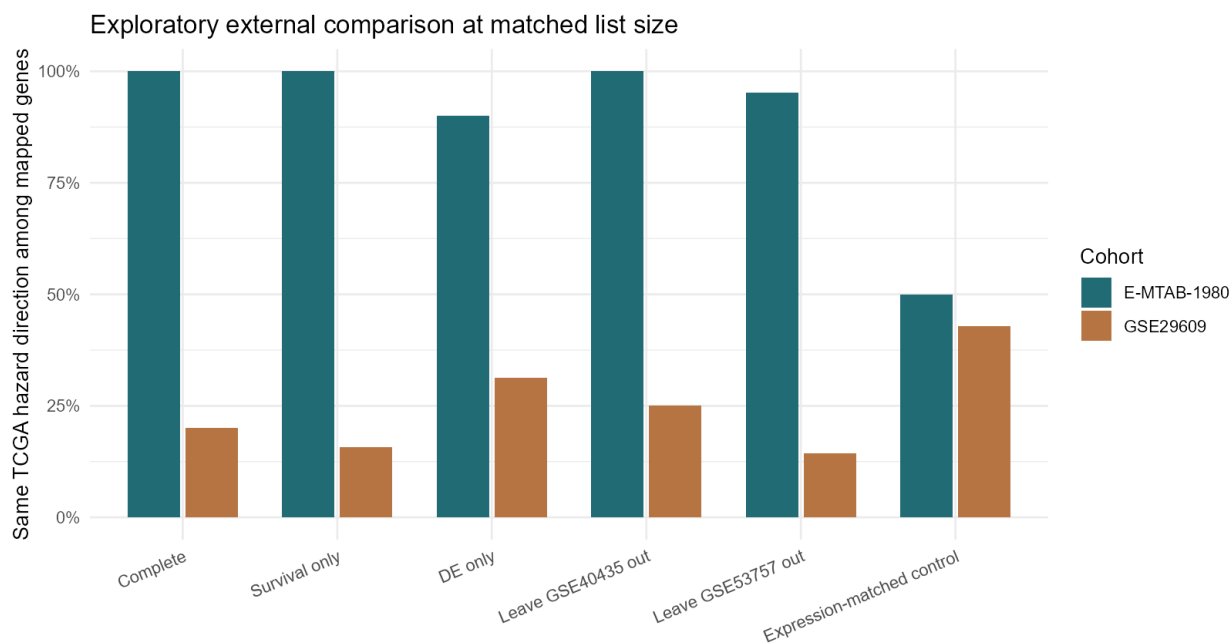


Figure 7: Same-direction replication among mapped genes for six equal-sized lists in two previously inspected cohorts. Different platform coverage and event counts limit comparison.

## External survival does not copy TCGA

Shaded rows are DDC and TCIRG1. GSE29609 has 17 deaths; E-MTAB-1980 has 23.

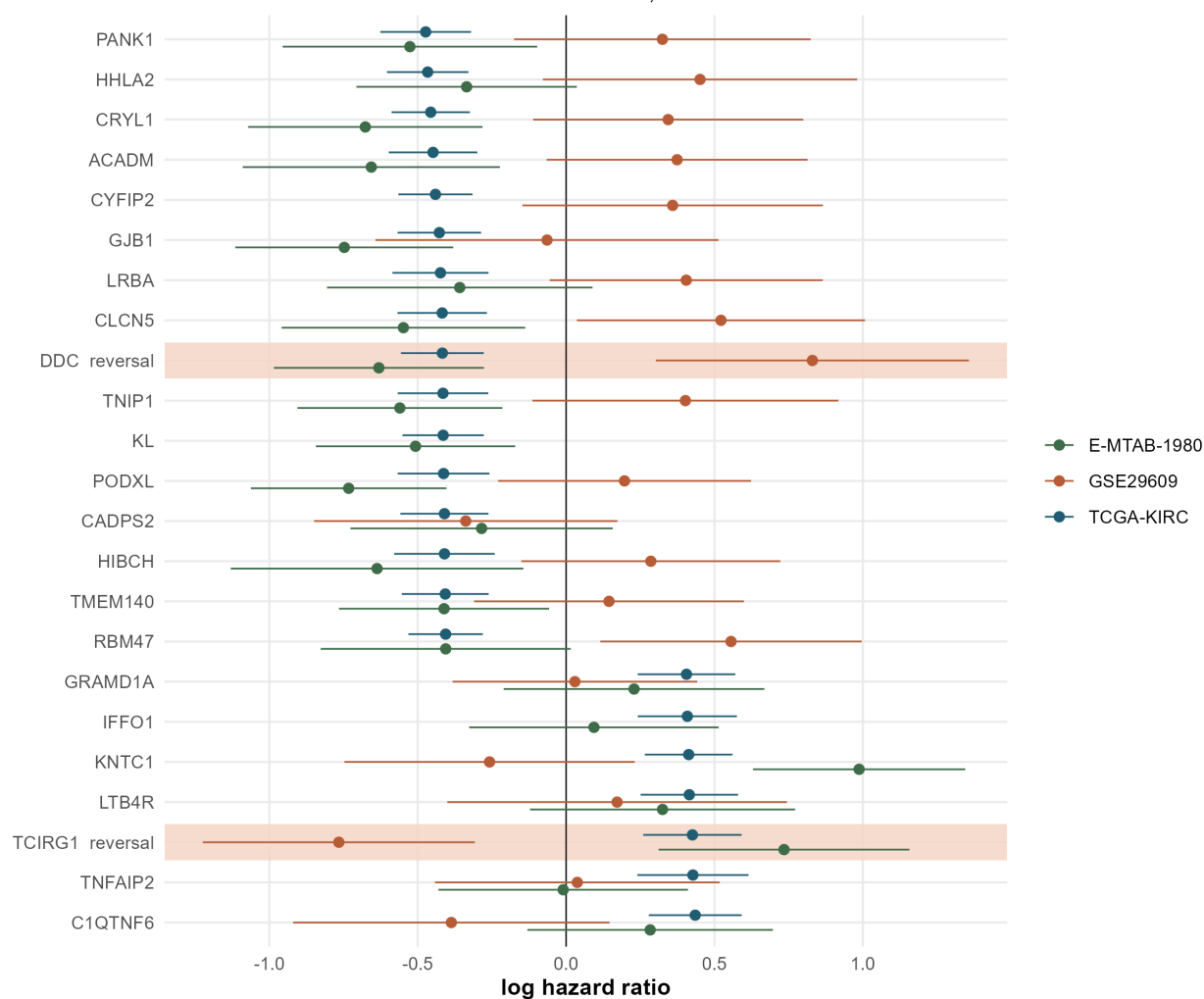


Figure 8: Log hazard ratios for the high-confidence genes in TCGA-KIRC, GSE29609 (17 deaths), and E-MTAB-1980 (23 deaths). Shaded rows are the DDC and TCIRG1 reversals. A shared direction is not an interval that excludes zero.

The two adjustments do not agree, so composition control is model-dependent. Consensus purity remains an inferred bulk estimate rather than a tumor-cell measurement.

### Composition adjustment of the 23 candidates

Triangles lose FDR support after marker-score adjustment. The line is 0.05.

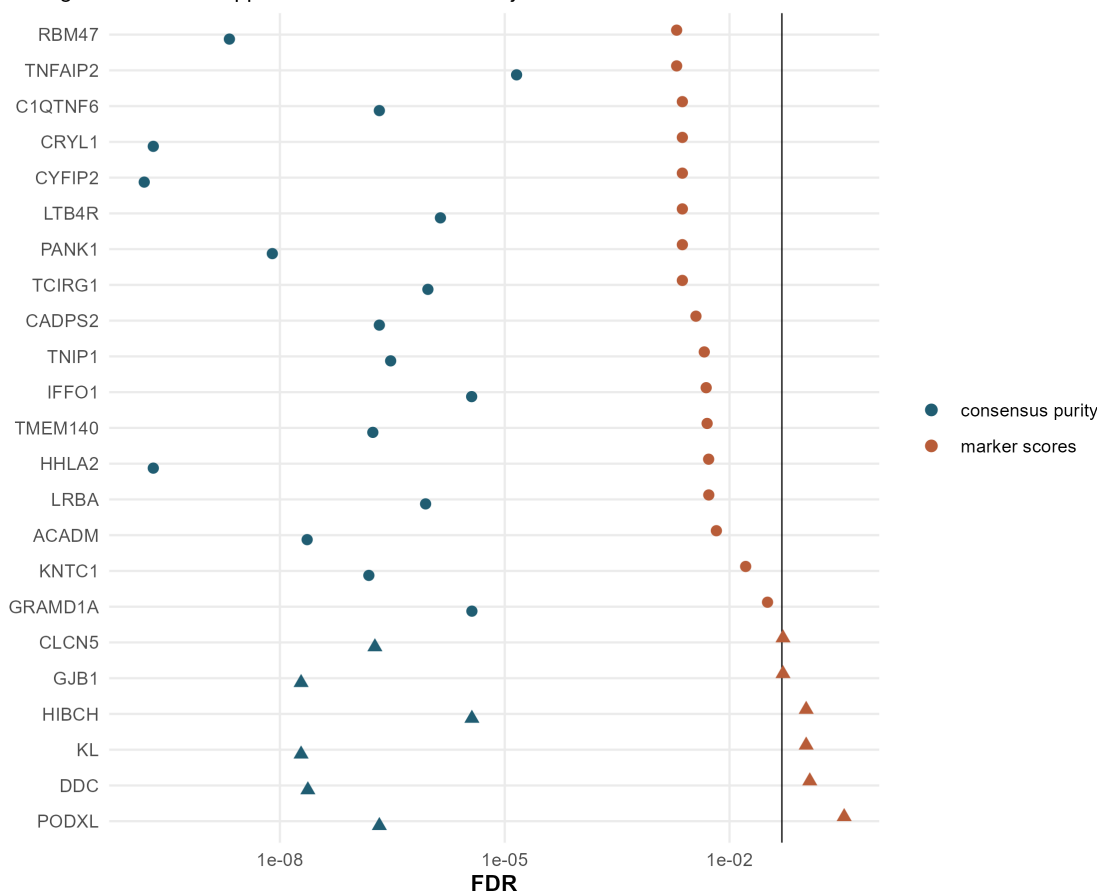


Figure 9: FDR after marker-score adjustment and after consensus-purity adjustment. Triangles mark the six candidates that lose FDR support under the marker scores. The vertical line is 0.05.

After correction for 23 tests, 0 nonlinear-expression comparisons and 0 expression-by-time comparisons had FDR below 0.05. Normal-tissue HPA profiles do not identify ccRCC tumor-cell origin.

### 3.4 Spatial sampling and randomized treatment

TRACERx linked 186 primary regions from 73 patients, including 16 deaths. A fixed event-stratified subset of 39 patients matches the size of GSE29609 and isolates region choice. Across genes, median agreement with the TCGA hazard direction was 98.5% (interquartile range 76.6 to 100.0) when those patients were fixed and only the chosen region varied, versus 86.3% (interquartile range 77.8 to 93.1) when both patient membership and region varied. Each smaller subset had 9 deaths. That event count makes this a sampling-variation check. It cannot attribute GSE29609's reversals to spatial sampling alone.

The CheckMate 025 RNA subset included 250 patients (120 nivolumab and 130 everolimus); all 23 candidates were assessed. No expression-by-treatment interaction survived FDR correction for overall or progression-free survival in either the randomized or limited-adjustment model. Nominal interactions remain exploratory and do not establish treatment selection.

## TRACERx sampling variation

Each draw keeps one region per patient. Both subsets have 39 patients and 9 deaths.

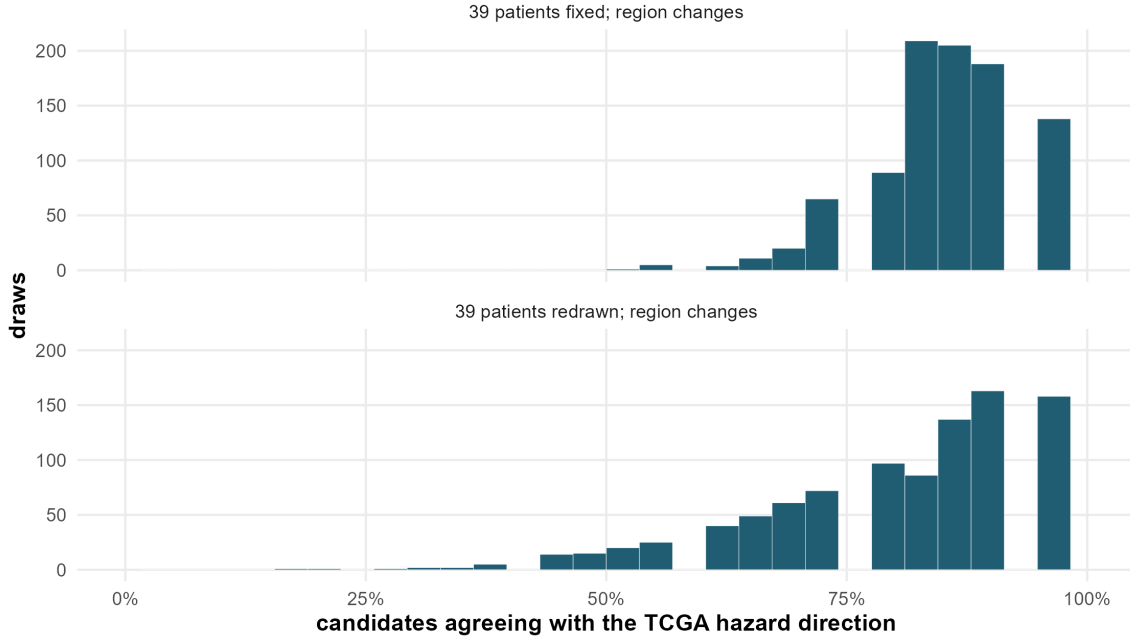


Figure 10: Fraction of high-confidence candidates that keep the TCGA hazard direction in 1,000 TRACERx draws. The upper panel holds the 39 patients fixed. The lower panel redraws which 39 patients enter. Both have nine deaths.

## 4 Discussion and limitations

The funnel fails as a one-gene selection procedure. The broader reproducible transcriptomic space still contains held-out prognostic ranking information. Counting each patient once, then repeating differential expression and candidate selection inside held-out folds, changed concordance by 0.0039, and the conditional patient-bootstrap interval crossed zero. That is a failure of the prediction criterion. It is not evidence that the 23 associations are null, and it is not a deployed signature. The ridge concordance increment of 0.0257 cleared 0.01 on the same splits. The penalty was training-only, the Brier interval still crossed zero, and the comparison was specified after the one-gene rule, so it does not replace the failed test. The TRACERx median of 98.5% is not near-complete agreement: the interquartile range across genes runs from 76.6% to 100.0% even when the 39 patients stay fixed. With 9 deaths, those draws cannot explain the GSE29609 reversals. Marker-score and consensus-purity adjustment also disagree, so a composition result from one of them is not a composition result from both.

Correcting patient identity changed the shortlist from 27 to 23 genes. The four genes that leave under every alternate aliquot rule show that the edge of the high-confidence set is sensitive to how the eight extra aliquots are handled. The unpaired GEO gate did not remove the 23. The permutation null selected a gene in 0 of 200 simulations, so scrambled training survival did not pass the high-confidence gate. That count is not a test of the concordance increment. ClearCode34's held-out increment was larger than the one-gene funnel increment, and its conditional interval still included zero and values below 0.01. Conditional coefficient precision and single-gene holdout performance did not translate into a prespecified gain for the complete training-selected procedure: its mean concordance increment was below 0.01, its conditional patient-bootstrap interval crossed zero, and gene identity varied between training folds. This is a failure of the stated prediction criterion, not evidence that every selected association is null. The same-split comparison locates that failure more narrowly. Removing the tumor-normal gate or the survival gate, and adding only the resulting gene, did not improve held-out ranking. Keeping every training-fold reproducible gene in a ridge

model did. The concordance gain cleared 0.01 with an interval above zero, while the Brier interval still included a worse score and the calibration slope stayed near the clinical value. The eligible expression set is therefore not empty of ranking information. The one-gene rule did not carry that ranking gain into held-out patients. The ridge fit is still not a locked signature: the penalty and the gene set were re-estimated in each training fold, GEO evidence stayed fixed, and the patient bootstrap did not refit that selection. The matched-list external comparison likewise failed to show a directional-replication advantage over survival-only ranking. None of these results supports a deployed multigene panel.

This is a retrospective public-data analysis. GSE29609 has only 17 deaths, E-MTAB-1980 has 23, and both were inspected before this reanalysis; no untouched external patient cohort is available. The matched-list interval resamples genes even though co-expression can make their directional results dependent, so it is descriptive rather than a confirmatory confidence interval. The 39-patient TRACERx subsets have nine deaths and do not recreate GSE29609’s platform or case mix. Consensus purity and bulk marker scores are imperfect composition measures; HPA profiles normal tissue, and tumor single-cell, protein, and functional validation are unavailable. The post hoc CheckMate 025 RNA subset and everolimus comparator do not validate predictive effects in current treatment settings. RNA associations cannot establish a mechanism or therapeutic target.

## Data and reproducibility

Public source identifiers, the pinned TRACERx commit, cached-input hashes, and package versions are recorded in the repository. The original retrieval dates for cached inputs are unknown. Core result counts and estimates are generated from result tables by `analysis/25_paper_numbers.R` into `paper/results_macros.tex`; the pipeline and PDF build are separate reproducible commands. This manuscript is an internal AI-assisted development draft for independent verification before submission.

## Ethics, funding, and AI assistance

This secondary analysis used public de-identified data and involved no new recruitment or intervention. No external funding was reported. OpenAI Codex assisted with the audit, code, and development manuscript; the student researcher must independently verify every statistical, scientific, and citation claim before any submission.

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